

Media Literacy

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Article 1

As a resident of a backcountry area of Southern California, the author of this lab has experienced planned power outages lasting multiple days. These outages are implemented by the power companies to reduce the chance of wildfires caused by downed power lines during strong wind events. In an effort to be more prepared in the future, the author started researching backup power systems. The two most common systems are gas-powered generators and large lithium batteries. The article below is one read during the process of determining which to buy. Read the article and answer the questions that follow.

Review: The Fast-Charging Power Station That Blew Up Kickstarter
September 20, 2019 | By Berne Broudy

Found at: <https://gearjunkie.com/ecoflow-delta-electric-generator-review>

EcoFlow's Delta 1300 battery powers your work and play tools emissions-free. And it claims to recharge faster than any other battery-based generator. No wonder it's nearing \$2 million on Kickstarter.

Brands like Goal Zero and Enerplex brought the lithium-ion battery pack into the main-stream during the past decade. But while its large battery systems have begun to compete with gas-powered generators, most consumers continue to use internal combustion for large-scale home use.

Recharging is one of the major hurdles of large batteries capable of powering tools and appliances. Most of the systems require a full day or more to recharge. Enter the EcoFlow Delta 1300 battery generator.

Live now on Kickstarter, it handles daily tasks and emergency power needs with one breadbox-size unit. It packs enough power to run shop vacs, table saws, microwaves, and computers. It can even give an electric car an extra 6-7 miles of power.

And it can recharge from stone dead to 80 percent in just one hour.

Small and portable enough to take into the woods or in an (electric) adventure rig, this battery can give you juice anywhere you can haul it.

Eco-Flow Delta Review

The 30-pound Delta 1300 has six AC outlets, two USB-C PD ports, and four USB outlets. A large LCD screen tells you how much battery the 100-cell lithium-ion bank has left by percentage and hours.

Push a button, turn it on, plug in a dead cellphone, and a fully charged Delta will tell you it has 99 hours of power remaining.

Plug in a 13-inch MacBook Air, and the readout bounces between 57 and 85 hours of power remaining. Because devices draw power inconsistently, numbers will fluctuate. So Delta tells you in real time how much juice it has left.

Unlike gas generators, which require annual maintenance, EcoFlow claims the Delta requires nothing and will hold its charge for a year. And while gas generators emit carbon monoxide and can be dangerous to use in a closed space, the Delta's only emissions are a quiet hum and a little bit of heat. So you can plunk the Delta onto your kitchen counter to run a microwave or coffee machine without worry.

Delta takes 2 hours to recharge plugged into a wall and has solar-charging capability. EcoFlow estimates the lifecycle to 800 charges. By contrast, Goal Zero's Yeti 1400 takes 25 hours to charge. It's also twice the weight, twice the price, and claims a lifecycle of 500 charges.

In my test at home, we ran every tool we had and charged every device: circular saws, table saws, shop vacs, computers, phones, even our refrigerator.

We were able to fully drain the battery during normal use only when we plugged in a full freezer trying to cool its contents from 14 degrees Fahrenheit down to zero. The battery lasted at least 20 hours — we woke up the next morning to it needing a recharge.

EcoFlow Electric Generator: How It Works

Delta's surprising power comes from a proprietary battery management system that, according to EcoFlow CEO Eli Harris, maximizes the power storage efficiency. It collects each cell's temperature and power status and tweaks the charging current and voltage for the safest, fastest charging.

EcoFlow also developed a charging platform that allows alternating current (AC) from a wall outlet to direct input into Delta's inverter, increasing its charging power at the same time. "By passing through the inverter directly, we can increase charging speed to more than 10 times the traditional AC-to-DC adapter cable," Harris said.

Recharge the Delta from a wall socket, carport, or solar panel. It plugs into the wall with the same cord you'd use to plug in a computer — no specialized, device-specific power brick required. So you don't have to worry about misplacing your charger.

And there's no fiddling with the unit when you want to recharge from a solar panel. All direct current (DC) power supplies below DC 60V — via an adapter, solar panel, or DC car output — go into a single input port.

EcoFlow Delta Battery

Delta manages rugged construction without a heavy, bulky, full-steel casing. Its Tesla-inspired housing uses high-grade aluminum and steel for strength and structural rigidity.

Impact-absorbing plastic, protective rigid metal plates, and four aluminum pillar reinforcements complete the packaging and ready Delta to withstand the hazards of a job site, garage project, or overland adventure.

Delta raised \$1.5 million in its first week on Kickstarter, and it's well on its way to surpassing \$2 million. We are impressed with its power, versatility, quick charge time, and compact size. If you support Delta before the campaign closes on October 19, you'll receive a lifetime battery warranty.

EcoFlow Delta 1300 Electric Generator

Weight: 30 lbs.

Ports: 6 AC, 2 USB-C PD, 4 USB

Shelf life: 12 months

AC output: 1,600 W (surge 3,100 W)

Charge time: 1.7 hours

Battery type: Lithium-ion

Price: \$1,400 (\$899 early bird buy-in)

Questions

1. Is this product the best solution to the problem of needing power backup during the annual fire season in California?
2. Whether you believe it is or not, list at least 3 reasons why it could be.
3. Do you see any drawbacks to this product?

Article 2

Now read a second article and answer the questions that follow.

5 Things to Know About Portable Power Stations

These pricey battery-powered devices are the only type of generator that can be used indoors safely

By Haniya Rae

April 12, 2019

Retrieved from: <https://www.consumerreports.org/portable-power-stations/portable-power-stations-things-to-know-battery-inverter-generators/>, 1/17/2020

Since generators emit carbon monoxide, they require that you take critical safety measures, including running the device outside, at least 20 feet away from any structure.

But in an age when we can charge our smartphones with a battery pack that fits inside a pants pocket, shouldn't there be a simpler way to restore power in the wake of a storm? Or, say, power a campsite without the constant hum of a gas-fueled generator?

Such is the promise of portable power stations, also known as battery-powered inverter generators. Essentially, they're oversized rechargeable batteries—about the size of a countertop microwave oven—that you plug into a typical 110-volt outlet to top off.

When duty calls, you can safely run a portable power station inside, since it doesn't generate any emissions. They have enough capacity to power a few small appliances for a short time. With a host of different outlets (standard 120v outlets, USB ports, and DC chargers), you can use the station to charge electronics, too. And the units often come with portable solar panels, to add more charging capabilities and extend runtime.

"These generators have no fumes and all of the models we tested made virtually no noise," says test engineer Dave Trezza, who oversees generator testing at Consumer Reports. "But, if these power stations go dead and you're unable to use your solar panels, you can't recharge them. You can't just use another gallon of gas."

We see models from brands, including Goal Zero, Humless, K2, and Kohler in our portable power station ratings. Some companies, like Goal Zero, market these portable power stations as perfect for apartment preparedness during storms (as opposed to single-family homes with a yard that can accommodate a generator).

How We Test Portable Power Stations

In our labs, CR test engineers evaluate five key measures to rate portable power stations: runtime, power delivery, power quality, ease of use, and noise.

To test runtime, we run a constant 300-watt load to simulate powering a TV and a few lights. We also hook each battery up to a side-by-side refrigerator to see how long it lasts. The best model in our tests powered the fridge for 44 hours on one charge (the worst only managed for 13 hours). For power delivery, or how well a model can maintain voltage

when tasked with different loads, we use a variety of devices, including a ½-horsepower submersible pump and a 10,000-BTU air conditioners.

We also judge noise output and found that, as a category, these batteries run quietly: All the models we tested earned an Excellent score for noise.

Below, our experts share some pros and cons on using portable power stations. See how the most popular models fared in our ratings, and check our generator buying guide to compare portable power stations to other types of generators.

1. They Can't Deliver Nearly as Much Power as Gas Generators

As with their gasoline-fueled counterparts, portable power stations require a transfer switch should you wish to power things such as your furnace, overhead lights, or any thing else in your home that's hardwired.

But while a recreational inverter generator would probably keep the TV and a few lights on for 8 to 13 hours on one tank of gas, you'd see anywhere from 3 to 9 hours of power, under the same circumstances, with a portable power station.

And you won't be able to run, for instance, your power-guzzling well pump.

2. They Take Awhile to Charge

In our tests, most of these models require hours of charging (typically overnight) to provide you with a full battery and max runtime.

So—assuming you fully charged the battery before a predicted weather event—a portable power station could give you hours of electricity to run a refrigerator or another essential appliance.

But Trezza notes that once the battery is dead, if you're without power and minimal sun, there's no way to recharge.

3. Charging With Solar Panels Can Be Iffy and Lengthy

If you're dealing with an outage or you're otherwise off the grid, charging the power station via solar panels is your only option—and that's provided you have good sun and no obstructions.

In our tests, we found that the solar panels can add to the runtime, but that might only amount to an extra hour or two of power with larger appliances.

4. They're Not All That Portable

Portable power stations are about the size of an average microwave oven, but they're fairly heavy—most in our tests weigh more than 80 pounds.

That means you'll likely need an extra pair of hands to lift one into the trunk of your car. Some of the models come with wheels, but not all wheels are large in size, which makes rolling them across a lawn difficult.

5. They Don't Come Cheap

The portable power stations we tested cost between \$1,500 and \$3,500. And our best portable gas generator costs less than the worst portable power station.

Before buying one of these power stations, consider if you might be better served by a portable generator. In the event of a outage, you can continue to add fuel, and portable generators typically provide enough power to keep larger appliances running. Check out our buying guide on generators to learn which type might best suit your needs.

Questions

Answer the following questions about article 2.

1. Do you believe that the product described in the first article is the best solution to the need of the author of this lab for backup power?
2. What are the positives to the battery backup solution in comparison to the gas-powered generator? Give at least three.
3. What are the drawbacks to the battery backup solution in comparison to the gas-powered generator? Give at least three.

Consider both articles as you answer these questions.

4. Did your answers about the benefits and drawbacks of the battery-powered product change after reading the second article? If so, how? If not, why did the additional information not influence you?
5. Who were the authors of the two articles? What, if anything, do you know about each of them?
6. From the articles and information provided about them, what can you tell about the credibility of each of these authors. What do you know, if anything, about how believable they might be?
7. Can you learn anything about the credibility of each of the articles from their URLs? Try to go beyond the dot extension (.org, .com).
8. What do you think is the purpose of each of the articles?
9. Do you think that the articles are current enough to be helpful?
10. Which article did you find most helpful? Would it have been enough to read just that one article?

11. Answer this question just for yourself: When was the last time that you read something that came from a differing opinion or perspective from your current belief – this could be political, socio-economic, religious, ethnic, gender, physical ability etc. just as a reality check for the validity of your view?

Bibliography